



Risk Analysis Ready Player Pi

Attachment 2.3

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Introduction

WHAT IS RISK ANALYSIS AND WHY

1. Risk Analysis

HOW ARE ALL PARAGRAHF STRUCTURED HERE

1.1 Project / Team

1.2 Database DREJER

- Structured/ Unstructured
 - Scoreboards
 - Usernames

1.3 Server/Docket MALTHER

- Multiple 2 player games
 - Threads
 - Game Rooms
- TCP

1.4 GUI

GUI-1: Incorrect GUI state handling between menus and game states

The GUI for both players contains multiple screens and layers such as username setup, main menu, room setup and more. Incorrect transitions between these menus could lead to users being stuck in a menu or entering the wrong menu.

Mitigation:

Use a GUI state machine to ensure consistent transitions between application states. Additionally, ensure that a “Back” button or navigation option is always available, allowing users to return to the previous menu if an incorrect state or navigation error occurs.

GUI-2: Username validation and storage errors

Errors using duplicate usernames or incorrect file handling could prevent usernames from being stored correctly.

Mitigation:

Input validation and safe file handling for storing usernames locally.

GUI-3: GUI desynchronization between clients due to communication delays

Since the system involves two players with separate GUIs, delayed server responses can cause the two interfaces to display outdated or inconsistent information relative to the expected game state. This could lead to confusion during gameplay.

Mitigation:

Use asynchronous communication to prevent the GUI from blocking while waiting for server responses and allow continuous user input.

GUI-4: Limited Graphical performance on RPi hardware - **Måske ikke relevant?**

The Raspberry Pi has limited graphical processing abilities, which may lead to slow rendering / reduced responsiveness.

Mitigation:

Optimize GUI rendering and minimize unnecessary updates.

GUI-5: Poor usability or unclear user instructions

If the interface is not intuitive, or any instructions are unclear, users may misunderstand how to navigate and use the system.

Mitigation:

Clear UI feedback and simple interface elements

1.5 GyroPi

GYRO-1: Lack of smooth movement of the ball on the Communicator Player's interface

If the gyroscope data from the Performer RPi is processed inefficiently, the ball movement on the Communicator's screen may appear unsmooth / uneven. This could reduce gameplay experience, as it may make communication and coordination difficult.

Mitigation:

Apply smoothing to gyroscope data and optimize update rate for visual ball movement on Communicator Player's GUI.

GYRO-2: Ball position desynchronization during gameplay

Due to communication delays between the two players' RPis (see GUI-3), the ball position on the Communicator's interface may not perfectly match the Performer's current action. This could lead to incorrect guidance during gameplay.

Mitigation:

Design the game to tolerate small delays by smoothing ball movements, as well as design a map that minimizes these issues.

GYRO-3: Difficulty controlling the game using the Game Controller

It might be hard for Performer Players to accurately control the ball, which could make the game unintentionally difficult.

Mitigation:

Adjust sensitivity of ball movement and scaling of gyroscope data. Additionally, provide visual feedback to Performer Player that indicates the current tilt, direction and magnitude, helping the player understand their current movements.

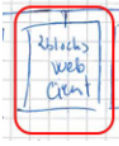
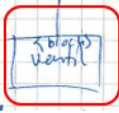
1.6 Game NotePi: Ida og Marie

- Recognizing tones
- Background noise
- Playing multiple tones at the same time
- Instrument /app with keyboard

SWOT analysis

	Strengths	Weaknesses
Opportunities	- Modular capabilities	- Dependent on very low latency
Threats	- Hardware malfunctions - Noices.	

Nedbringelse af riscisi

Minimumskrav?	Hvilke muligheder findes?	Argumenter for / imod?	Kandidat	Forsøg
Én bruger Tekst er ok Skal køre på RPI	PHP .NET Core + Angular Rå HTML HTML + JS HTML + JS + Websockets	Ifølge K er der med S begrænset mulighed for... Bla bla bla	Rå HTML	Lav en simpel hello world html side. Gerne som kører på Rpi
Godkendt til fødevarer Min/maks tryk	Ventil X Ventil Y Hjemmelavet ventil, 3dprint + motor	Bla bla bla	Ventil Y	Lav simpel styring af ventil med lab forsyning og Analog Discovery. Mål strømforbrug

Lav simpel journal per undersøgelse, eller notér grundigt i logbog. –her er guf til analysen i rapporten!

Risk mitigation tool:

1. Envision the risk items
2. Evaluate the risk items
 - a. Probability
 - b. Consequence
3. Make risk mitigation / contingency for each risk item

Identification

Project impact

- Schedule, resources, people skills, estimates, tool
 - o Problem with tool, missing important competances

Product impact

- Requirements, quality, performance, components, technology

Risk matrix

2. Risk Matrix FÆLLES

Likelihood

Harm severity

	Minor	Marginal	Critical	Catastrophic
Certain				
Likely				
Possible			Dropping the RPi.	
Unlikely				High latency.
Rare				

Potential term list

- **Catastrophic:** Permanent disability, total loss of equipment
- **Critical:** accident level injury resulting in hospitalization, permanent partial disability, damage to equipment
- **Marginal:** injury causing lost workdays, reversible moderate environmental impact, minor accident damage level
- **Minor:** injury not causing lost workdays, damage less than a minor accident level

3. Risk Analysis Proccesseseseses

TIME MANAGEMENT"

PROBLEMER MED HVORDES EMFETODER

NEW PROJECT MODEL - SCRUM

4. Risk Matrix



Consequence/ Likelyshood	Negligible	Minor	Moderate	significant	Severe
Almost certain					
Likely					
Possible					
Unlikely					
Rare					